Vikesh Kumar

Generative AI — LLM Engineering — Agentic AI — LangChain — LangGraph — RAG [
+91-9935477768 | vikeshkumar965166@gmail.com | LinkedIn | GitHub

Professional Summary

Aspiring Generative AI Engineer specializing in Large Language Models (LLMs), multi-agent systems, and Retrieval-Augmented Generation (RAG). Built production-style AI applications using LangChain, LangGraph, FastAPI, and Groq APIs, including automated research assistants, document intelligence systems, and machine learning prediction services. Passionate about Agentic AI, workflow orchestration, and scalable AI applications, with a strong problem-solving background demonstrated through 400+ solved DSA problems.

Education

ABES Engineering College, Ghaziabad, Uttar Pradesh 2024 – 2028 (Expected)
Bachelor of Technology in Computer Science and Engineering
CGPA: 7.8 / 10 **(Till 4th Semester)**

Skills

Programming Languages: Python, C++, SQL
Generative AI LLM Engineering: Large Language Models (LLMs), Generative AI, Agentic AI, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, Prompt Engineering, LLM Workflows, Context Engineering, Groq API
Vector Databases Retrieval: FAISS, ChromaDB, Semantic Search, Vector Embeddings, Document Retrieval
Machine Learning Data Science: Scikit-learn, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Model Deployment, Predictive Analytics
Natural Language Processing (NLP): Text Processing, Tokenization, Text Chunking, Information Extraction, Document Summarization
Frameworks Libraries: Streamlit, Pandas, NumPy, Matplotlib, Seaborn
Developer Tools Platforms: Git, GitHub, LangSmith, Jupyter Notebook, Kaggle
Core Computer Science: Data Structures Algorithms, Object-Oriented Programming (OOP), Operating Systems

Projects

Multi-Agent Research Assistant GitHub Repository
Built a multi-agent AI research platform using Python, LangChain, LangGraph, Groq API, Tavily Search, BeautifulSoup, and Streamlit to automate information gathering, analysis, and report generation.
Orchestrated four specialized AI agents (Search, Reader, Writer, and Critic) using LangGraph to enable autonomous task decomposition and multi-step reasoning workflows.
Integrated web search and scraping pipelines to collect, process, and synthesize information from multiple online sources in real time.
Developed LLM-powered workflows using Groq-hosted LLaMA models for research summarization, content generation, and response evaluation.
Implemented a feedback-driven validation loop where a Critic Agent reviewed outputs for factual consistency, completeness, and quality before final report generation.
Designed and deployed an interactive Streamlit application enabling end-to-end research automation and structured report creation.

LLM-Powered Academic PDF Intelligence Platform Live Demo ;—; GitHub Repository
Built and deployed an LLM-powered document intelligence application using Python, LangChain, Groq API, and Streamlit to transform academic PDFs into structured study materials.
Developed an end-to-end AI pipeline for PDF ingestion, text extraction, preprocessing, chunking, prompt engineering, and automated content generation.
Implemented RecursiveCharacterTextSplitter-based chunking to efficiently process long-form documents while preserving contextual relevance and response quality.
Leveraged Groq-hosted LLaMA models to generate concise summaries, detailed explanations, key concepts, and exam-oriented study notes from unstructured educational content.
Engineered custom prompting workflows to automatically generate bullet-point summaries, 2-mark conceptual questions, and 7-mark descriptive examination questions.
Optimized inference workflows to achieve 5–8 second response times for large academic documents, improving usability and user experience.

ML-Powered Student Placement Prediction Platform **Live Demo ;—; GitHub Repository**

- Built and deployed an end-to-end machine learning application to predict student placement probability using academic performance and skill-based attributes.
- Performed data preprocessing, exploratory data analysis (EDA), and feature engineering to improve data quality and model performance.
- Trained and evaluated Scikit-learn models, achieving 85
- Designed a production-style ML workflow covering data preparation, model training, evaluation, deployment, and inference.

ACHIEVEMENTS

- **Solved 400+ DSA problems** across competitive platforms: **CodeChef (255+)**, **LeetCode (52+)**, and **HackerRank (100+)**.

LICENSES & CERTIFICATIONS

- **Introduction to Generative AI** — Google Cloud (Simplilearn SkillUp), Aug 2025 Credential Link
- **Python (Basic)** — HackerRank, Feb 2026 Credential Link